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Management Science

Publication details, including instructions for authors and subscription information:
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To cite this article:
Patrick Behr, Darren J. Kisgen, Jérôme P. Taillard (2016) Did Government Regulations Lead to Inflated Credit Ratings?. Management Science

Published online in Articles in Advance 05 Dec 2016

. http://dx.doi.org/10.1287/mnsc.2016.2615

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MANAGEMENT SCIENCE
Articles in Advance, pp. 1–22
ISSN 0025-1909 (print) | ISSN 1526-5501 (online)

https://doi.org/10.1287/mnsc.2016.2615
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Did Government Regulations Lead to Inflated Credit Ratings?

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Securities and Exchange Commission (SEC) regulations in 1975 gave select rating agencies increased market power by increasing both barriers to entry and the reliance on ratings for regulations. We test whether these regulations led to ratings inflation. We find that defaults and negative financial changes are more likely for firms given the same rating if the rating was assigned after the SEC action. Furthermore, firms initially rated Baa in the post-regulation period are 19% more likely to be negatively downgraded to speculative grade than firms rated Baa in the pre-regulation period. These results indicate that the market power derived from the SEC led to ratings inflation.

Keywords: credit ratings; SEC; default risk; NRSROs; capital markets regulation; regulatory licensing
History: Received November 12, 2015; accepted July 5, 2016, by Amit Seru, finance. Published online in *Articles in Advance* December 5, 2016.

1. Introduction

Credit ratings assess the creditworthiness of corporate bonds, but the quality of these ratings has been put in question. While credit rating agencies (CRAs) can play a useful role by more efficiently gathering information on firm default likelihoods and providing that information to dispersed investors, they have also been criticized for missing major default events (e.g., mortgage-backed securities and Enron). Two key factors that affect credit rating agencies are (1) the competitive environment in the ratings industry and (2) the use of ratings in rules and regulations.[1] In this paper, we evaluate a key event that materially altered both of these factors. As such, our paper contributes directly to the current debate surrounding credit ratings by examining the impact of regulations that both decreased rating competition and increased ratings-based regulations.

In June of 1975, the use of ratings in rules and regulations was significantly expanded when the Securities and Exchange Commission (SEC) issued new rules that established bank and broker-dealer capital requirements based specifically on ratings (Rule 15c3-1). The competitive landscape among CRAs was also significantly altered at this time, as the SEC deemed that only ratings from certain rating agencies (Moody's, Standard and Poor's (S&P), and Fitch) could be used in regulations. These agencies were defined for the first time as nationally recognized statistical ratings organizations, or NRSROs. This watershed event certified the value of ratings provided by these NRSROs to companies and investors, thereby giving these rating agencies new significance in the marketplace. The introduction of the NRSRO status came at a time when the environment for credit rating agencies was becoming increasingly competitive with the rise of several new entrants in the 1970s threatening the dominating role of the big three. The barriers to entry established by the NRSRO designation, as well as the expansion of the use of ratings in regulations, effectively limited competition and gave these designated rating agencies a significant competitive advantage. The objective of this paper is to investigate the effects of these changes empirically.

Our main tests evaluate whether the assignment of ratings differs following the introduction of the SEC's new NRSRO and credit rating-based capital requirement rules in June 1975. Specifically, we first compare default likelihoods for firms rated before the new regulations to the default likelihoods for firms whose ratings were initiated with the same rating in the

[1] For example, the Dodd–Frank Act calls for investigations into both of these issues.

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post-SEC-certification period. If the new rules have no effect on ratings, we would expect that, for example, a firm rated Baa before the rule change should have, on average, the same probability of default as a firm rated Baa after the rule change. Instead, we find that, conditional on a given rating, firms have a greater likelihood of default if they were rated in the post-SEC-certification period relative to those rated before June 1975. We also examine changes in financial performance, proxied by Altman's Z-score and find similar results. Over our evaluative period going up to December 1987, firms of a given rating in the post-period suffer, on average, a more pronounced drop in financial performance than firms assigned with the same rating in the pre-period. Together, these results indicate that the regulatory events of 1975 led to ratings inflation. To our knowledge, this is the first paper to show that the increased use of ratings in rules and regulations through the introduction of NRSRO status is at least partially responsible for ratings inflation in the credit rating industry. This implies, among other things, that recent initiatives to expand the number of certified rating agencies will have a positive impact on ratings objectivity.

We conduct two additional tests to estimate the size of this ratings inflation. First, we artificially deflate post-SEC-certification ratings to the point at which the ratings would appear to not be inflated based on realized default probabilities. Second, we compare actual ratings given after SEC certification to predicted ratings for those firms based on their fundamentals (such as leverage) using a ratings prediction model derived from ratings and firm fundamentals in the pre-period. In both of these tests, we find that ratings are inflated by approximately three-fourths to one full letter grade (e.g., firms rated Baa should be rated Ba).

We next conduct tests that examine the investment grade (IG)/speculative grade (SG) boundary. If rating agencies use the incentive of regulatory certification to attract new clients, we expect them to focus on firms that might properly be rated speculative grade by giving them an investment grade rating (Baa or better). We test this conjecture by examining whether firms given an investment grade rating after SEC certification are more likely to be subsequently downgraded to speculative grade than firms that are rated investment grade before SEC certification. We find that they are, and that the effect is concentrated in firms given a Baa (investment grade threshold) rating. Firms given a Baa rating in the post-period are over 19% more likely to be subsequently downgraded to speculative grade relative to firms given the same Baa rating in the pre-period. These results provide further evidence consistent with the SEC conferring a regulatory-based competitive advantage to NRSRO rating agencies, thus giving them the opportunity to focus on new business creation at the expense of some ratings integrity.

With any empirical tests relying on one event to provide exogenous variation, one should assess the potential for other contemporaneous changes to confound the interpretation of the results. One such concern would be any changes in macro conditions around the time of the change in regulations.[2] We perform several falsification tests to rule out empirically the potential confounding influence of the macro environment at the time of the event. Another potential confounding event is Standard and Poor's switch from investor pay to issuer pay in July 1974. However, we conduct additional tests that indicate this event does not drive our findings.

One other potential implication of the new SEC regulations is that ratings information content might deteriorate. Rating agencies may rationally exert less effort generally if they know they have a captive customer base, and ratings would therefore become less precise. We find no evidence consistent with this implication. In tests examining rating upgrade and downgrade frequencies, only the frequency of rating downgrades increases following the regulation. This result is consistent with ratings inflation but not with a decrease in information content. Furthermore, default likelihoods appear to maintain ordinal integrity following the regulations (e.g., an A-rated firm still has a lower default likelihood than a Baa firm, etc.). Thus, the SEC regulation leads to a general upward shift in ratings levels, but differences in ratings are still informative. Ratings inflation benefits institutional investors that are constrained by regulations, but losing accuracy in general would make ratings less attractive for nonconstrained investors. We conclude that rating agencies use higher levels of ratings to attract clients, but maintain ordinal ranking to appeal to investors who continue to seek information from ratings.

[2] Our sample period contains a recession in 1973–1974 and another recession in 1981–1982. The latter occurs during our evaluative window and impacts all sample bonds equally. The 1973–1974 recession occurs during our pre-SEC-certification window. Rating agencies claim to "rate through the cycle," implying ratings levels are unaffected by changes in macro conditions that are considered normal phases of the business cycle. If a typical recession does not impact subsequent default risk, our results will be unaffected. But if recessions increase default risk and bonds are given the same rating as in a boom phase, bonds rated in a recession will have a higher risk of defaulting simply because they are rated in a recession. Given that the 1973–1974 recession occurs *before* the SEC event, this would introduce a bias *against* finding post-SEC-certification ratings inflation. Consistent with this reasoning, our results are stronger when we exclude all bonds from the pre-SEC-certification period rated between November 1973 and March 1975, the recession phase according to National Bureau of Economic Research data.

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Our paper is part of a growing literature that shows how competition in the rating industry and the use of ratings in rules and regulations impact credit ratings, both theoretically and empirically (see, e.g., Becker and Milbourn 2011, Bongaerts et al. 2012, Bolton et al. 2012, Bar-Isaac and Shapiro 2013, Manso 2013, Opp et al. 2013). Work by Blume et al. (1998), Alp (2013), and Baghai et al. (2014) shows that ratings have become more stringent over time. Given the increased competition among CRAs over that same time period, their evidence suggests that competition does not lead to rating inflation (consistent with our results). However, Becker and Milbourn (2011) find a positive relation between competition among CRAs (as measured by Fitch's market share over time) and assigned credit ratings during the period 1995–2006.[3] Recent work by Bae et al. (2015) argues that the results of Becker and Milbourn (2011) seem to be driven by regulated industries. Opp et al. (2013) provide a theoretical model evaluating the impact of regulations based on ratings and find that "relative to the equilibrium without regulations, the rating agency has an incentive to rate more firms highly" (p. 48). Our results provide some empirical support for their theoretical predictions. Finally, our results are also in line with theoretical work by Bar-Isaac and Shapiro (2013). They show that rating agencies' incentives to provide quality ratings are lower when the cost of reputation loss is lower, which arguably is the case when regulatory power is greater and competition is lower.

2. Regulatory Background of Ratings, Theoretical Considerations, and Hypothesis Development

2.1. Regulatory Background

On June 26, 1975, the SEC started to use credit ratings issued only by nationally recognized statistical rating organizations in its Rule 15c3-1, which determined net capital requirements for brokers and dealers (SEC Release No. 34-11497). Beginning with that event, the use of NRSRO ratings in rules and regulations became widespread as other regulators followed suit, adopting the notion of NRSRO status for their own rating-based regulations (see Bethel and Sirri 2016).

These new rules contained two key parts. The first was the new regulation for broker-dealers. The SEC adopted Rule 15c3-1 that set forth broker-dealer "haircut'' requirements that were a function of the credit ratings of those securities. Through this rule, the SEC applied a net capital rule whereby credit ratings became the basis for determining the percentage reduction in the value ("haircut") of bonds held by broker-dealers for the purpose of calculating their capital requirements. The second is the establishment of NRSROs, which gave select rating agencies significant power with respect to rules and regulations, thus creating significant barriers to entry for competitors. These two regulatory changes combined represent one of the most significant events in the history of ratings regulation. Corporations also took notice of these developments. Recent research suggests that corporations adjust capital structure and investment decisions based directly on concerns for these regulations, confirming the importance of these regulations to rating agency clients (see, for instance, Kisgen 2006, 2009; Begley 2015).

The competitive landscape in the credit rating industry was evolving rapidly in the 1970s, with at least five rating agencies other than the big three entering the market around that time. These new entrants threatened the dominance of Moody's, S&P, and Fitch. Duff and Phelps analyzed fixed income securities dating back to the 1930s, and started providing letter designations in the mid-1970s using an issuer pay model. They started making the ratings public in 1980, received NRSRO status in 1982, and their rating business was purchased by Fitch in 2000. McCarthy, Crisanti and Maffeir, Inc. was founded in 1975, but did not receive NRSRO status until 1983, and was purchased by Duff and Phelps in 1991 (which in turn was purchased by Fitch). Thomson Bankwatch started its rating services in 1974, and Canadian Bond Rating Service was founded in 1972 (and was later acquired by S&P in 2000). Finally, Dominion Bond Rating Service was founded in 1976 but did not receive NRSRO status until 2003.

Despite the emergence of new competitors, when the SEC established the use of NRSROs in 1975, only Moody's, S&P, and Fitch were grandfathered in. In the 25 years that followed this SEC certification in 1975, only four new firms were given NRSRO status, and all four ended up merging or being acquired by the big three agencies such that only the original three had NRSRO status 25 years later. The process of obtaining NRSRO status has been contentious, with many firms complaining that they have been unfairly excluded from this select club (see Kisgen and Strahan 2010). The NRSRO status granted in 1975 led to significantly reduced competition for the three largest rating agencies. As White (2010) argues, "the Securities and Exchange Commission crystallized the centrality of the three rating agencies in 1975" (p. 213).

[3] Relatedly, Griffin and Tang (2011) and Griffin et al. (2013) show that rating agencies adjusted upward their ratings for securitized products in the run-up to the 2008 financial crisis. Their evidence is supportive of catering theories whereby Moody's (and S&P) inflated their ratings when they faced competition from one another on collateralized debt obligations (CDOs). In particular, rating agencies seem to have provided more favorable ratings for larger issuers of CDOs to generate more fee-based revenues in the future (see He et al. 2012).

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The change in regulations in 1975 was largely unexpected as well. The SEC awarded the NRSRO status through a "no-action" letter system, such that a firm could request to use a CRA for regulatory purposes, and so long as the SEC granted the "no action" letter for it, the CRA's rating would qualify for regulatory purposes (for details, see Bethel and Sirri 2016). No specific criteria or definition was outlined, but the most important factor was that the rating agency be "nationally recognized" in the United States as a credible provider of ratings. In that regard, the new rule appears to have been somewhat informally determined.[4]

2.2. Theoretical Considerations and Hypothesis Development

The 1975 regulations restricted competition and provided value to NRSRO-designated ratings, as both firms and investors had to rely on NRSRO ratings to meet their regulatory requirements. It is important to note that while we analyze the joint effect of the decrease in competition and increased regulatory importance of ratings following the 1975 event, our empirical setup does not allow us to disentangle the two effects, and hence we cannot make statements about which of the two effects prevails.

Regarding competition, Manso (2013) argues that increased competition can lead to lower ratings when rating agencies want to be more accurate than their competitors. Outside of the credit ratings literature, several additional papers document that reducing competition reduces quality, both theoretically and empirically.[5] However, Bolton et al. (2012) show that an increase in competition might lead to more rating shopping and hence higher reported ratings.

Regarding the second component of the 1975 regulations, the use of ratings in rules and regulations, Opp et al. (2013) provide a theoretical model evaluating the impact of regulations based on ratings and find that "relative to the equilibrium without regulations, the rating agency has an incentive to rate more firms highly" (p. 48); that is, following the implementation of rating-based regulations, their model would predict an overall increase in ratings (ratings inflation). Partnoy (1999) proposes that with the "regulatory license" granted in 1975, rating agencies "no longer need to generate valuable information to sustain their reputations for quality over time" (p. 698). Bolton et al. (2012) also show that "lower reputation costs give CRAs incentives to inflate investment quality" (p. 109). Furthermore, Bar-Isaac and Shapiro (2013) provide a model that indicates a credit rating agency has less incentive to provide quality ratings when the cost of reputation loss is smaller. The 1975 regulations reduce the cost of reputation loss for NRSRO-designated rating agencies by creating a captive clientele, and thus create more incentives for them to inflate ratings according to their model. As a result of the 1975 regulations, rating agencies can more easily cater to their clients' needs by inflating ratings and thus capturing the regulatory rents granted to them through the NRSRO-designation process.

Additionally, Rule 15c3-1, detailing broker-dealer haircut requirements, provided specific benefits to firms that had one of the top four rating levels from at least two certified rating agencies, i.e., an investment grade rating (Baa rating for Moody's). All debt instruments rated at or above this threshold qualify for IG holdings, which extends significantly the pool of potential investors and lowers the cost of capital for firms that can obtain this investment grade rating.[6] Consequently, firms benefit significantly from obtaining an investment grade rating (even if the ratings are later readjusted downward), and this effect provides the incentives for NRSRO-designated rating agencies to disproportionally inflate ratings around the IG cutoff.

Based on these theoretical considerations, we hypothesize that the three rating agencies designated as NRSROs become more entrenched following the new SEC rules in 1975, and therefore are more likely to inflate their ratings after SEC certification. The null hypothesis is that despite the increased regulatory power and lower threat of competition following these changes, NRSRO rating agencies continue to provide unbiased ratings. Since rating agencies also need to protect their reputation to some extent to retain clients and maintain their NRSRO status, rating agencies might want to provide even more stringent ratings to establish their credibility. Given this and considering that a reduction in competition could also lead to less ratings inflation (as in Bolton et al. 2012), the null could be rejected in favor of ratings deflation.

In our main tests, we compare the predictive power of Moody's bond credit ratings at the end of May 1975, the pre-SEC-certification period, to the predictive power of Moody's bond credit ratings that were issued between July 1975 and December 1978, the

[4] Under pressure from Congress, the NRSRO granting process was overhauled and formalized with the passage of the Credit Rating Agency Reform Act in 2006 (for more details, see Bethel and Sirri 2016).

[5] For example, Matsa (2011), Bloom et al. (2015), Cooper et al. (2011), and Mussa and Rosen (1978).

[6] See, for instance, Kisgen (2006), Kisgen and Strahan (2010), and Begley (2015). As Kisgen and Strahan (2010) find, at least one-third of the yield difference between the investment grade and speculative grade is due to regulations. Furthermore, Bongaerts et al. (2012) find evidence consistent with more ratings shopping from firms around the investment grade cutoff.

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post-SEC-certification period.[7] To test the hypothesis that NRSRO-designated rating agencies inflated their ratings after SEC certification, we compare the realized default likelihood for bonds rated in the pre-SEC-certification period to the realized default likelihood for bonds rated in the post-SEC-certification period. Under the inflation hypothesis, we expect that, for any given credit rating level, the likelihood of default will be higher in the post-SEC-certification period.

Finally, following the theoretical consideration that ratings inflation should be most valuable for firms near the investment/speculative grade threshold (e.g., Opp et al. 2013), we also examine the impact of the regulations on ratings for firms near the investment grade boundary. We hypothesize that after the change in regulations, rating agencies had particularly strong incentives to inflate ratings around the investment grade threshold because this is where the regulatory value of ratings is the greatest.

3. Data and Summary Statistics

To implement the empirical tests, we first collect all credit rating information on bonds that are issued until December 1978 from Moody's Corporate Default Risk Services (DRS) database. We restrict our sample to U.S. corporate issuers of nonconvertible bond/debentures ratings. For instance, we exclude sovereign ratings, structured finance ratings, and pooled investment vehicles ratings. Within all corporate issuers, we focus our analysis on firms classified as "banks," "finance," "industrial," "transportation," and "utilities." The DRS database provides us with detailed data on each issuer. It also gives us rating data at the time of issuance and all subsequent credit events, including defaults and credit rating changes, over the lifetime of each bond in our sample.

The Altman Z-score tests require accounting and market data on bond issuers. As such, we hand-collect Center for Research in Security Prices (CRSP) and Compustat identifiers for each issuer and then merge these accounting and market data to the Moody's data set. The resulting data set contains credit, accounting, and market value variables for each bond in our sample. In particular, these data allow us to compute the Altman Z-score as an alternative credit risk outcome measure for every firm over time.[8] All accounting items are taken from Compustat, and the market value of equity is taken from CRSP.

We focus on valid long-term ratings issued by Moody's. The rating classification goes from "Aaa" (highest credit quality) to "C" (lowest credit quality).[9] The letter rating is coded numerically in our data, with the highest rating "Aaa" getting the lowest score (1) and the lowest rating "C" getting the highest score (9). Table 1 provides an overview of the sample characteristics.

In panel A of Table 1, we give the number of issues within each rating category per end of May 1975 for all outstanding issues rated before the SEC certification (pre-NRSRO status), and for all new issues rated between July 1975 and December 1978 for the post-SEC-certification period. Overall, the sample contains 2,496 observations (1,991 pre-NRSO and 505 post-NRSO). Most bond issues have investment grade ratings, i.e., ratings of Baa or better, both in the pre- and post-periods. This result is not surprising given that we analyze a period before the vast increase of the junk bond market in the 1980s. The majority of bonds are issued by industrial firms (∼60% of the sample), with bonds of public utility firms being the second most important category in our sample (∼20%) and the rest being split among banking, financials, and transportation.

Panel B of Table 1 provides averages by credit rating category for four firm-level characteristics known to affect a firm's creditworthiness. In particular, these four firm characteristics are used by Blume et al. (1998) in their rating level prediction analyses. The first variable is the book value of total assets in USD millions, a proxy for firm size. The second is leverage, the third is pretax interest coverage, and the fourth is operating margin. As would be expected, higher-rated firms are (1) larger, exhibit (2) less leverage, display (3) higher interest coverage, and have (4) higher operating margin than lower-rated firms.

We compute these averages separately for pre-period bond issues versus post-period bond issues. There are no clear patterns across the two time periods in terms of firm size and interest coverage. Both operating margin and leverage seem to be higher on average in the post-period issues. In our prediction tests, we more formally evaluate whether pre- and post-firms differ on fundamentals.

4. Main Empirical Results

4.1. Default Likelihood and Financial Performance

4.1.1. Graphical Evidence. We begin our analysis with a graphical representation of default frequencies

[7] This window length allows us to capture the fact that the NRSRO-based competitive advantage for the top three rating agencies was potentially only gradually recognized following the first "no action" letter in June 1975.

[8] The Altman Z-score is computed as Z-score = 1.2 ∗ Working Capital/Total Assets + 1.4 ∗ Retained Earnings/Total Assets + 3.3 ∗ Earnings Before Interest and Taxes (EBIT)/Total Assets + 0.6 ∗ Market Value of Equity/Book Value of Total Liabilities + 0.999 ∗ Net Sales/Total Assets.

[9] We only look at full letter grade ratings. Moody's introduced a more refined rating classification, the so-called "notching," in April 1982. Before April 1982, there was no distinction within a given credit rating category.

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Table 1 Descriptive Statistics

Panel A: Sample size and industry distribution

	Rating distribution			Industry distribution (%)				
Rating category	#Pre	#Post	Total	Banking	Finance	Industrial	Public utility	Transportation
Aaa	390	97	487	7.8	7.6	77.4	4.3	2.9
Aa	326	94	420	6.4	19.1	52.6	19.8	2.1
A	752	167	919	2.5	11.9	54.2	26.7	4.7
Baa	330	69	399	0	3.8	56.1	33.1	7.0
Ba	124	22	146	0	0.7	50.6	34.3	14.4
B	57	56	113	0	0.9	80.5	1.8	16.8
Caa-C	12	0	12	0	0	33.3	0	66.7
Totals	1,991	505	2,496	3.5	9.7	59.7	21.4	5.7

Panel B: Sample characteristics

	Observations		Assets		Leverage (%)		Interest coverage		Operating margin	
Rating category	#Pre	#Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Aaa	174	57	6,129	7,027	56.0	54.9	3.546	4.298	0.249	0.256
Aa	245	57	1,636	3,367	53.6	59.9	4.192	3.873	0.159	0.242
A	524	112	1,202	1,089	58.8	67.1	3.415	3.005	0.133	0.206
Baa	226	49	1,027	1,352	63.7	63.6	2.633	2.938	0.136	0.131
Ba	84	18	1,210	454	65.5	68.4	2.529	4.505	0.139	0.141
B	36	38	445	136	70.0	66.5	1.660	3.343	0.056	0.078
Caa-C	0	0	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Total/Average	1,290	331	1,340	1,406	58.7	64.6	3.194	3.349	0.1559	0.1913

Notes. This table provides descriptive statistics for our sample. The default sample comprises all debt issues in our default tests. The SEC established the nationally recognized statistical rating organization certification in June 1975. The sample is comprised of all bond issues with an existing rating from Moody's as of May 1975 (pre-NRSRO sample) as well as all newly rated issues from July 1975 to December 1978 (post-NRSRO sample). In panel A, we provide the number of observations by rating category and the industry distribution within our sample. In panel B, we provide the number of observations for which firm-level data are available, the median total assets (in million USD), leverage ratio, pretax interest coverage ratio, and operating margin for both the pre- and post-NRSRO-certification samples. For the pre-NRSRO sample, the values are taken as of year-end 1974, while the values for the post-NRSRO sample are taken as of the end of year in which the initial rating was assigned.

by rating category over the period running from July 1975 to December 1987 in Figure 1. We report separately the default frequencies for bond issues rated in the pre-SEC-certification period versus those rated in the post-SEC-certification period.[10] For ratings between Aaa and Aa, there are almost no defaults and, hence, no pronounced increase for post-SEC-certification-rated bonds. For all other rating categories, the graph shows a clear increase in the default frequency of bonds rated in the post-SEC-certification period. To complement this figure, we provide in Figure 2 the average Z-score change by rating category for issues rated by Moody's both before and after the SEC certification. The interpretation is consistent with Figure 1. There is a greater deterioration in financial condition within almost every credit rating category for the post-SEC-certification issues. Both figures are consistent with ratings inflation in the post-period.

4.1.2. Default Analysis. We now more formally test the relationship documented graphically above and analyze whether the NRSRO status granted to credit rating agencies in 1975 led to ratings inflation. Specifically, we test whether Moody's credit ratings differ in predicting defaults before and after the introduction of the NRSRO status. To test for rating inflation following the regulatory events of 1975, we regress realized defaults on a dummy variable indicating whether the rating was assigned in the pre- or post-SEC-certification period.

In these tests, we control for rating levels by including an indicator variable for each rating level (rating level fixed effects).[11] Controlling for the rating level in this test is crucial because otherwise a higher default likelihood of post-SEC-certification period ratings could simply be driven by comparing bonds of different risk classes. Within a given rating category, we predict a systematic bias in the predictive performance of post-SEC-certification ratings relative to pre-SEC-certification ratings if the new regulatory power

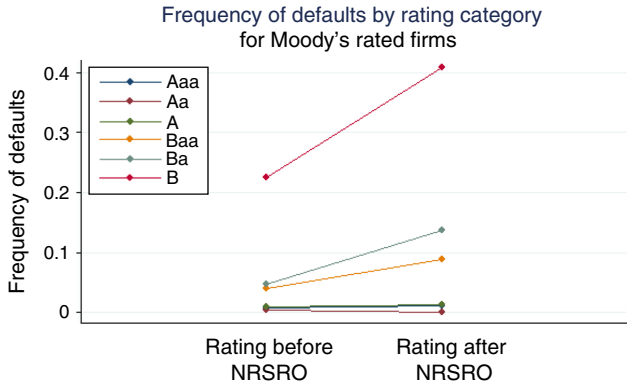
[10] There were no issues assigned a rating between C and Caa in the post-period; hence, we do not represent those categories in Figures 1 and 2.

[11] The inclusion of rating level fixed effects addresses the concern that a bond's default risk is a nonlinear function of the letter rating grade. We obtain statistically and economically stronger results if we include the numerical rating as a continuous variable instead.

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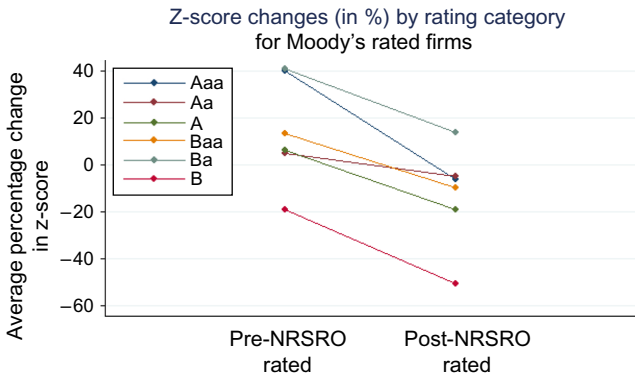
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Figure 1 **(Color online) Default Frequencies by Rating Category Before and After SEC Certification**



Notes. This figure presents the frequency of defaults by rating category for bond issues rated by Moody's both before and after the SEC certification (introduction of NRSRO status in June 1975). For the pre-SEC-certification issues, the ratings are taken as of May 1975. The ratings are taken at issuance for all issues after SEC certification. The *y* axis shows the frequency of defaults over the period running from July 1975 to December 1987 for pre-period issues, and from issuance date to December 1987 for all post-NRSRO bonds issued between July 1975 and December 1978.

Figure 2 **(Color online) Z-Score Changes by Rating Category Before and After SEC Certification**



Notes. This figure presents the average Z-score change by rating category for issues rated by Moody's both before and after the SEC certification (introduction of NRSRO status in June 1975). For the pre-SEC-certification issues, the ratings are taken as of May 1975. The ratings are taken at issuance for all issues after SEC certification. The *y* axis shows the average Z-score change (in percentage) up to December 1987. The starting value for the Z-score is taken as of year-end 1974 for pre-NRSRO issues, and as of the year-end prior to issuance for all post-NRSRO bonds issued between July 1975 and December 1978. We link the average changes in Z-score across the two subsets of bond issuers within each categories to provide a visual aid as to the general pattern observed across all ratings before and after SEC certification.

of rating agencies affects their incentives to provide unbiased ratings. Beyond the rating level, we also include industry fixed effects to control for any risk differences between industries not captured by the credit rating.

Most of our tests do not include firm-specific controls. We argue that any firm-specific factor should be subsumed in the rating because the credit rating is a risk predictor that is designed to summarize firm-specific risk factors. Moreover, while the firms in our sample may differ along any observable or unobservable dimension, such differences should be captured by the credit rating if the rating is a meaningful indicator of credit risk. This implies that, for instance, a Baa-rated bond in 1974 should have the same risk of defaulting as a Baa-rated bond in 1976, unless the predictive power of both ratings differs. That being said, we do show results of some tests with additional firm level controls, and our main findings are not affected. Our baseline regression is the following:[12]

$$\text{default}_i = \alpha + \beta * \text{post} + \gamma * \text{I\{rating-level\}}_i + \delta * \text{controls}_i + \text{error}_i. \quad (1)$$

The dependent variable default_i is an indicator variable taking on a value of one if a default of bond i occurs within a specified time period ("evaluative window") after the introduction of the NRSRO status; $post_i$ is an indicator variable taking on the value of one for all new issues rated between July 1975 and December 1978 and zero if the bond issue is from the pre-SEC-certification period; $\text{I\{rating-level\}}_i$ corresponds to the set of indicator variables for each credit rating level. Only the indicator corresponding to the credit rating of bond i will equal one, with all the others set to zero; controls_i include industry fixed effects controlling for the industry of the bond issuer and, in some specifications, firm-specific characteristics; error_i is the error term of the regression.

The rating level used in the regressions is the rating as of May 1975 for bonds issued in the pre-SEC-certification period and the initial rating of the bonds issued in the post-SEC-certification period. The use of the initial rating for the post-SEC-certification period bonds assures that we have the smallest possible time difference between the pre- and post-SEC-certification period issues. All results hold if we use the end of December 1978 ratings for the post-SEC-certification period instead. Standard errors are clustered at the issuer level in all regressions.[13]

For our main test, we have to define a time period (evaluative window) during which we record the number of defaults for both the pre- and post-SEC-certification period bond issues. For the baseline analysis, we record all defaults from July 1975 to end

[12] The estimation is done using ordinary least squares (OLS) despite the binary character of the dependent variable. Our main results hold if we use logit or probit instead. As an alternative specification framework, we estimate duration models later in this section to take into account censoring in the data.

[13] All results remain unchanged when we cluster on the industry level instead, using Moody's issuer industry classification.

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Table 2 Default Regressions

Panel A: Defaults as of year-end 1987				
	(I)	(II)	(III)	(IV)
Post-dummy	0.0239**	0.0264**	0.0345**	0.0314**
	[0.0103]	[0.0116]	[0.0142]	[0.0141]
Aa-dummy	−0.0064	0.0016	−0.0136	−0.0034
	[0.0087]	[0.0090]	[0.0177]	[0.0188]
A-dummy	0.0020	0.0092	−0.0087	−0.0036
	[0.0105]	[0.0114]	[0.0205]	[0.0208]
Baa-dummy	0.0400**	0.0462**	0.0270	0.0305
	[0.0181]	[0.0190]	[0.0254]	[0.0261]
Ba-dummy	0.0546**	0.0602**	0.0369	0.0356
	[0.0260]	[0.0258]	[0.0357]	[0.0348]
B-dummy	0.2590***	0.2530***	0.2186***	0.2006***
	[0.0504]	[0.0506]	[0.0657]	[0.0660]
Caa to C-dummy	0.1632	0.1564	−0.0351	−0.0829*
	[0.1080]	[0.1086]	[0.0388]	[0.0464]
Asset size (million USD)			−0.0510	−0.0469
			[0.1368]	[0.2157]
Leverage			0.0385	0.1238
			[0.0534]	[0.0808]
Interest coverage			0.0563	0.0669
			[0.1012]	[0.1078]
Operating margin			−0.0310	−0.0247
			[0.0519]	[0.0603]
Observations	2,496	2,496	1,621	1,621
Adjusted R^2	0.113	0.119	0.095	0.105
Industry fixed effects	No	Yes	No	Yes

of December 1987.[14] The regression coefficient β in regression model (1) indicates whether bonds have a different default frequency depending on whether they were issued before or after SEC certification, controlling for the respective rating level in a nonlinear way. If SEC certification leads to rating inflation, we would expect β to be positive and significant.

Our baseline regression results are shown in panel A of Table 2. Column (I) of panel A shows a positive and highly significant coefficient for *post-dummy* of 0.0239, implying that the default frequency of post-SEC-certification bonds is 2.39 percentage points higher than that of pre-SEC-certification bonds after controlling for credit rating fixed effects. This effect is both statistically and economically significant considering that the (unconditional) pre-SEC-certification default frequency is only 2.16%. In columns (II)–(IV) we include sequentially industry fixed effects and the four firm characteristics from Table 1 as additional control variables.[15] The results are unchanged in these alternative specifications, although the sample size is smaller because of the lack of firm-specific information in some cases.[16] The coefficients for the different rating level indicator variables have to be interpreted relative to the omitted rating category Aaa. They have the sign and magnitude we would expect. In particular, bonds with higher numerical ratings (i.e., higher credit risk) show a higher likelihood of defaulting over time.

By recording defaults over the period from July 1975 until end of December 1987, we are in effect using different time windows over which we estimate the predictive power of ratings across the pre- and post-SEC-certification bond issues. Specifically, for bonds rated in the precertification period, our time window extends over a period of 12 years and six months. For bonds rated in the post-SEC-certification period, the window can be as short as nine years for bonds issued in December 1978. The use of different time windows may bias our results, but the bias is against finding inflated ratings for post-period-rated bonds, since the post-period-rated bonds have a shorter window during which to default.

We nonetheless run a second test in which we use time windows of same length. Although this

[14] In robustness tests below we show that all results hold if we choose longer or shorter time windows.

[15] Another important control variable is bond maturity. We show in the robustness tests section below that different bond maturities for pre- and post-period bonds cannot explain our findings.

[16] The reason why the sample size is reduced in these tests is that we were not able to find Compustat data for them. Many of these cases are linked to subsidiaries of larger publicly traded companies.

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Table 2 (Continued)

Panel B: Defaults over 10-year window
(Pre-period issues (June 1975–May 1985)/Post-period issues (December 1978–December 1988))

	(I)	(II)	(III)	(IV)
Post-dummy	0.0406***	0.0423***	0.0475***	0.0461***
	[0.0112]	[0.0122]	[0.0147]	[0.0145]
Aa-dummy	−0.0007	0.0067	−0.0005	0.0085
	[0.0036]	[0.0048]	[0.0066]	[0.0083]
A-dummy	0.0095	0.0162*	0.0022	0.0049
	[0.0070]	[0.0085]	[0.0119]	[0.0121]
Baa-dummy	0.0416***	0.0472***	0.0320*	0.0316*
	[0.0156]	[0.0167]	[0.0172]	[0.0178]
Ba-dummy	0.0479**	0.0522***	0.0287	0.0215
	[0.0205]	[0.0201]	[0.0250]	[0.0228]
B-dummy	0.2160***	0.2110***	0.1702***	0.1490***
	[0.0455]	[0.0454]	[0.0521]	[0.0519]
Caa to C-dummy	0.1727	0.1636	−0.0309	−0.0805**
	[0.1077]	[0.1083]	[0.0305]	[0.0379]
Asset size (million USD)			−0.1124	−0.0932
			[0.2125]	[0.1329]
Leverage			0.0605	0.1512**
			[0.0446]	[0.0716]
Interest coverage			−0.0142	0.0173
			[0.0623]	[0.0707]
Operating margin			−0.0327	−0.0231
			[0.0493]	[0.0580]
Observations	2,496	2,496	1,621	1,621
Adjusted R^2	0.102	0.107	0.091	0.105
Industry fixed effects	No	Yes	No	Yes

Notes. This table presents results for our baseline predictive default regressions. The dependent variable is an indicator variable taking on a value of one if a default of the bond occurs within a given evaluative window after the introduction of the NRSRO status. In panel A, the evaluative window over which a default can occur is from July 1975 to December 1987 for pre-NRSRO issues, and from the date of issuance to December 1987 for all post-NRSRO bonds issued between July 1975 and December 1978. In panel B, the time window is 10 years for both pre- and post-samples, running from July 1975 to May 1985 for the pre-NRSRO-rated sample and from December 1978 to December 1988 for the post-NRSRO newly rated issues. The *post-dummy* variable is an indicator variable for the post-NRSRO sample, i.e., for all new issues rated between July 1975 and December 1978. We include a rating dummy variable for each rating category. For each rating dummy, the variable takes a value of one if a given bond is assigned the corresponding rating level, as of May 1975 for bonds issued in the pre-SEC-certification period and at issuance for the bonds issued in the post-SEC-certification period. We omit the Aaa category such that the coefficients of the rating level dummies have to be interpreted relative to that category. Industry fixed effects are included in some of our specifications, as indicated in the table. Standard errors clustered at the issuer level are shown in brackets.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

test has the advantage of having a predictive window of identical length across pre- and post-SEC-certification bond issues, it has the disadvantage of having nonoverlapping years. Specifically, we compare defaults of pre- and post-SEC-certification issues 10 years after they were rated, which corresponds to the period 1975–1985 for pre-period-rated issues and 1978–1988 for post-period-rated bonds. If macro conditions between 1978 and 1988 differ considerably from those between 1975 and 1985, our results could simply reflect changing macro conditions over those nonoverlapping years. The results are displayed in panel B of Table 2. We find positive and highly significant coefficients for *post-dummy* in all four specifications. The magnitude of the economic effect is stronger than before. This finding reinforces the conclusions drawn from panel A of Table 2.

4.1.3. Z-Score Analysis. We expand our default analysis to a broader range of changes in credit risk by employing a widely used and accepted measure, the Altman Z-score (Altman 1968), as an alternative outcome variable. The Z-score has the advantage of being able to detect changes in credit risk for all observations, rather than only for those bonds that actually default. However, as the Altman Z-score cannot be computed for banks, they are not included in the analysis. This restriction and the lack of accounting data for some firms reduce the sample size to 1,063 observations. We run two different regressions. First, we estimate the following regression model:

$$\text{Z-score change}_i = \alpha + \beta * \text{post} + \gamma * \text{I}\{\text{rating-level}\}_i + \delta * \text{starting Z-Score}_i + \text{industry}_i + \text{error}_i. \quad (2)$$

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In this regression, *Z-score change*$_i$ is a dummy variable indicating whether the Altman Z-score of the issuer of bond i *decreased* over the evaluative window for bonds issued in the pre-SEC-certification period compared to bonds issued in the post-SEC-certification period. The *starting Z-score*$_i$ is the Altman Z-score of the issuer of bond i at year-end 1974 for the pre-SEC-certification bond issues that were issued until December 1974, and at year-end 1975 for the pre-SEC-certification bond issues that were issued between January and May 1975. For the post-SEC-certification bond issues, the starting Z-score is the Z-score at year-end of the year in which the initial rating was assigned for the bond.[17] The Z-score change is computed as the difference between the Z-score as of December 31, 1987, and the starting Z-score. We define our indicator variable relative to decreases in Z-score so as to reflect a worsening in credit risk. Doing so allows us to have the same interpretation as before for our coefficient of interest β: If the SEC certification led to inflated ratings, we would expect β to be positive and significant. The inclusion of the starting Altman Z-score controls for the fact that the initial Z-score may impact whether or not the Z-score drops during the sample period. The variable industry$_i$ corresponds to industry fixed effects for the issuer of bond i. The first column of panel A of Table 3 contains the baseline Z-score results for the time window until December 1987, and the second column contains the results for the 10-year window with nonoverlapping years.

Table 3 shows a positive and highly significant regression coefficient for the post-period indicator variable across every specification. This implies that firms that issued bonds in the post-SEC-certification period experienced decreases in their Altman Z-score more often than firms with bonds given the same rating in the pre-SEC-certification period, controlling for the starting Altman Z-score and the rating level in a nonlinear fashion. This result is in line with the default frequency results because it suggests that a deterioration of credit risk happened more often for firms whose bonds were rated in the post-SEC-certification period. The magnitude of the coefficient indicates that a decrease in Altman Z-score is approximately 15% more likely for post-period-rated bonds. This result is almost unchanged when we use the 10-year nonoverlapping time window in the second column.

In panel B of Table 3, we modify Equation (2) and regress the Altman Z-score *level* on *post-dummy* and controls instead. Once again, we transform the dependent variable and take its negative value such that a positive coefficient for β indicates a greater *drop* in Z-scores for post-period-rated firms. By controlling for the starting Altman Z-score, this regression essentially estimates the size of the change in the Altman Z-score for firms that issued bonds in the pre-NRSRO period versus firms that issued bonds in the post-SEC-certification period. We run this test for both the overlapping time window up to 1987 and the 10-year nonoverlapping time window. The results are displayed in panel B of Table 3. In both cases, we get positive and highly significant coefficients of similar magnitudes. Together, these results provide support for the conjecture that rating agencies inflated ratings following the change in ratings regulation and the

Table 3 Z-Score Regressions

Z-score as of:	End of 1987	After 10 years
Panel A: Indicator for drop in Z-score levels		
Post-dummy	0.1446***	0.1912***
	[0.0526]	[0.0653]
Starting Z-score	0.1716***	0.1488***
	[0.0230]	[0.0200]
Observations	1,066	1,105
Adjusted R^2	0.203	0.194
Rating level fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Panel B: Negative Z-score levels		
Post-dummy	0.3058***	0.4183***
	[0.1079]	[0.1188]
Starting Z-score	−0.5866***	−0.5693***
	[0.0573]	[0.0508]
Observations	1,066	1,104
Adjusted R^2	0.480	0.455
Rating level fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

Notes. This table presents the results for our baseline Z-score regressions. In panel A, the dependent variable is an indicator variable taking on a value of one if the Altman Z-score of a firm *decreases* between year-end 1974 and December 1987 for pre-NRSRO issues for the issues rated until December 1974, and between year-end 1975 and December 1987 for pre-NRSRO issues for the issues rated between January 1975 and May 1975 (and, respectively, year-end after the debt issuance and December 1987 for post-NRSRO issues) in the first column, and for 10 years after year-end 1974 or 1975 (respectively, year-end after issuance for post-NRSRO issues) in the second column. The negative change in the Z-score dummy is defined to reflect an increase in credit risk over our evaluative period. In panel B, the table shows regression results with Z-score levels as the dependent variable. It is defined with a negative sign to also reflect an increase in credit risk over the evaluative period. *Starting Z-score* is the Altman Z-score of the firm as of year-end 1974 (respectively, year-end 1975 for the pre-period issues rated between January 1975 and May 1975) for the pre-SEC-certification bond issues and per year-end of the year in which the initial rating was assigned for the post-SEC-certification bond issues. All other variables are defined in Table 2 and included as indicated in the table. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

***Indicates significance at the 1% level.

[17] If the initial rating was assigned, for instance, in September 1975, the starting Z-score would be the Altman Z-score as of the year-end 1975; if it was assigned in September 1976, the starting Z-score would be the Altman Z-score per the end of 1976, etc. This way, the computation of the Z-scores always fully incorporates the newly issued bonds in the year in which the rating was assigned.

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restriction in competition induced by the introduction of the NRSRO status.

4.2. Firms Near the Investment Grade Threshold

In this section, we investigate whether firms for a given investment grade rating in the pre- or post-SEC-certification period had a higher likelihood of being downgraded to speculative grade. If SEC certification led to inflated credit ratings, the likelihood of being downgraded to speculative grade should be higher for the post-SEC-certification period issues. The regulatory incentives for being investment grade were increased following 1975; hence, rating agencies would have stronger incentives to inflate ratings and assign investment grade ratings even if the risk of the bond would not justify an investment grade rating (even if temporary, this provides benefits in the form of lower capital charges for investors). As rating agencies focus more on client generation, Ba-rated firms might be enticed to enter the bond market if given an inflated Baa rating. If this is the case, we expect that IG bonds issued in the post-period are associated with a higher likelihood of being downgraded to speculative grade afterward. We estimate the following OLS regression model to test this hypothesis:

$$\text{SG-downgrade}_i = \alpha + \beta * \text{post} + \gamma * \text{I\{rating-level\}}_i + \text{industry}_i + \text{error}_i. \quad (3)$$

In this regression, SG-downgrade_i is a dummy variable taking on the value of one if bond i was downgraded from investment to speculative grade at any given point in time during the evaluative period.[18] All other variables are the same as before. The coefficient β on *post-dummy* indicates whether bonds that were rated in the post-SEC-certification period had a higher likelihood of being downgraded to speculative grade in the predictive window. If the SEC certification led to inflated ratings, we would expect β to be positive and significant.

Table 4, panel A, provides summary statistics on the frequency of downgrades below the investment grade threshold before and after the regulatory change. The table displays that for each investment grade rating category, the downgrades to SG are much more pronounced for the post-NRSRO-rated bonds. Table 4, panel B, more formally tests whether post-period-rated bonds are more likely to be downgraded below the investment grade threshold than pre-period-rated bonds. The results in panel B show that investment grade-rated bonds that were issued after SEC certification had an almost 8% higher likelihood of being downgraded to speculative grade until December 1987 relative to pre-SEC-certification-issued bonds. This effect is significant at the 1% level. In the second column of Table 4, we see that the effect is even greater and again highly significant when we use the 10-year nonoverlapping time window. These results provide additional evidence of rating inflation after SEC certification.

In panel C of Table 4, we focus only on Baa-rated firms. Firms near the investment grade threshold have the most to gain from inflated ratings as regulations concentrate significantly at that level. Thus, we would expect firms near the threshold to be especially prone to enticement from rating agencies of preferential ratings. Tests examining the transition from investment grade to speculative are a natural environment in which to consider this possibility. If firms are given a Baa rating when they in fact deserved a Ba rating or worse, these firms will be the most

[18] In case a bond was downgraded more than once to speculative grade, it is only counted as one downgrade in the analysis.

Table 4 Downgrades to Speculative Grade Regressions

Panel A: Descriptive statistics

Percentage of downgrades to SG as of:	End of 1987 (%)		After 10 years (%)	
	Pre	Post	Pre	Post
Aaa-rated bonds	1.03	7.22	1.03	7.77
Aa-rated bonds	8.28	7.45	6.13	10.53
A-rated bonds	11.04	20.96	7.18	21.74
Baa-rated bonds	22.46	48.53	20.31	50.75

Panel B: All investment grade-rated firms

Downgrades to SG as of:	End of 1987	After 10 years
Post-dummy	0.0777***	0.1192***
	[0.0273]	[0.0284]
Observations	2,219	2,219
Adjusted R^2	0.084	0.095
Rating level fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes

Panel C: Only Baa-rated firms

Downgrades to SG as of:	End of 1987	After 10 years
Post-dummy	0.1905**	0.2461**
	[0.0939]	[0.0952]
Observations	393	392
Adjusted R^2	0.116	0.127
Industry fixed effects	Yes	Yes

Notes. This table presents results for the downgrades to speculative grade regressions. Panel A provides statistics of the percentages of downgrades to SG by rating category. Panel B gives the regression results for all investment grade-rated bonds, while panel C focuses only on Baa-rated bonds. The dependent variable takes on the value of one if an investment grade bond is downgraded from investment grade to speculative grade over a given time frame. The time frame runs from June 1975 to year-end 1987 in the first column, and for 10 years as of May 1975 for pre-NRSRO-rated bonds and for 10 years as of December 1978 for post-NRSRO newly rated bonds in the second column. All explanatory variables are defined in Table 2. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

** and *** indicate significance at the 5% and 1% levels, respectively.

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Table 5 Magnitude of Ratings Inflation

Defaults per end of 1987:	Baseline	One rating category	Two rating categories
Panel A: Post-SEC-certification issue ratings *deflated* by one and two rating categories			
Post-dummy	0.0264**	−0.0047	−0.0344**
	[0.0116]	[0.0124]	[0.0144]
Observations	2,496	2,495	2,492
Adjusted R^2	0.119	0.128	0.128
Rating level fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Panel B: Pre-SEC-certification issue ratings *inflated* by one and two rating categories			
Post-dummy	0.0264**	−0.0063	−0.0432
	[0.0116]	[0.0158]	[0.0305]
Observations	2,496	2,009	1,589
Adjusted R^2	0.119	0.131	0.121
Rating level fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Notes. This table gauges the magnitude of ratings inflation for bond issues rated in the post-SEC-certification period. We estimate the same predictive default regressions as in Table 2, panel A, whereby the dependent variable is an indicator variable taking on a value of one if a default of the bond occurs until December 1987. The first column replicates the results of column (II) of Table 2, panel A (baseline). In panel A, we deflate all post-SEC-certification issue ratings by one (second column) and two (third column) rating categories; i.e., we change a Baa rating to Ba, etc. In panel B, we inflate all pre-SEC-certification issue ratings by one (second column) and two (third column) rating categories; i.e., we change a Ba rating to Baa, etc. All other control variables are defined in Table 2. Rating level and industry fixed effects are included in all specifications. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

**Indicates significance at the 5% level.

likely to be downgraded to speculative grade at a later point. The results of panel C of Table 4 confirm this conjecture. Baa-rated firms in the post-period are especially likely to be downgraded to speculative grade when compared with Baa-rated firms before the SEC action. The likelihood of a downgrade to speculative grade is 19% higher for firms rated Baa in the post-period compared to Baa-rated firms in the pre-period.[19] Together, these results confirm that Baa firms are given favorable ratings in the post-period to keep them, at least temporarily, above the investment grade threshold.

4.3. Magnitude of Ratings Inflation

The effects we document above are consistent with inflated ratings for post-SEC-certification-issued bonds because of rating agencies' changed incentive structure. In this section, we document the size of the ratings inflation. We test this by artificially deflating all post-SEC-certification bond ratings by one and two rating categories and running our baseline default regressions using the deflated ratings variable as a control. We replace all post-SEC-certification bond ratings by their rating minus one (one grade below actual; i.e., all bonds with an Aaa rating receive an Aa rating, all bonds with an Aa rating receive an A rating, etc.), while we leave unchanged the pre-SEC-certification bond ratings. In these tests, the C (Ca) ratings are excluded when we deflate by one (two) rating categories as these cannot be further deflated. The regression specifications are the same ones of the baseline default regressions. Panel A of Table 5 displays the results.

The first column reports the baseline result from Table 2, panel A. In the second and third columns, we replace the observed bond rating by the artificially deflated post-SEC-certification bond ratings by one and two rating categories, respectively. After a one rating category deflation, *post-dummy* is zero and not significant anymore; it turns negative and significant following a deflation of two rating categories in the third column. This result implies that the post-SEC-certification bond ratings are inflated by approximately one full rating category; i.e., bonds rated Baa after SEC certification would exhibit no rating bias if they were rated Ba instead, etc.[20]

In panel B of Table 5, we provide the symmetric test by artificially inflating all pre-SEC-certification bond ratings by one and two rating categories and run the same tests as above.[21] We find results similar to those

[19] While the effect for Baa-rated firms only is more than twice as large as for all investment grade-rated firms (19% versus 7.8%), the difference is not statistically significant because of the large standard errors associated with our estimates. The same holds when Baa-rated firms are compared to A-rated firms only. The result for the latter is much smaller, but the difference is not statistically significant because of the large standard errors.

[20] Besides providing insight about the size of the bias, this test also lends credibility to our econometric approach.

[21] The sample size is smaller as we remove Aaa ratings and both Aaa and Aa when we inflate by one (respectively, two) rating categories the pre-period bond ratings.

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from panel A, whereby *post-dummy* is close to zero and not significant after a one rating category inflation and turns negative (albeit not statistically significant) following a two rating category inflation. Taken together, the results in this section provide support for post-SEC-certification ratings inflation and suggest that the size of the inflation is substantial.

4.4. Assessing Rating Bias from Predicted Ratings

An alternate way to assess ratings bias is to examine whether ratings given after the introduction of the NRSRO status and the change in regulation appear inflated based on firm financials compared to the financials of similarly rated firms in the pre-period. For example, in the summary statistics in panel B of Table 1, the average A-rated firm has a leverage of 58.6% in the pre-period compared to 64.5% in the post-period, suggesting that A-rated firms are of lower credit-worthiness in the post-period based on this important variable for ratings determination. However, ratings are based on observable and unobservable risk characteristics, so no model relying solely on observables can definitively predict ratings. Hence, we believe examining defaults as the primary outcome variable provides the most direct methodology for assessing whether ratings are inflated. Nevertheless, in this section we examine whether ratings also appear inflated based on firm-specific characteristics.

To determine whether ratings appear inflated based on firm-specific characteristics, we compare actual ratings for firms in the post-period to predicted ratings for those firms based on a ratings model derived using pre-period data. Specifically, we estimate a model for ratings predictions by implementing a probit regression of pre-period firms' ratings on their asset size, leverage, pretax interest coverage, and operating margin. These four variables are four of the most important drivers of a firm's ratings (Blume et al. 1998, Kamstra et al. 2001, Kisgen 2006). Furthermore, we add cash flow volatility, computed annually as the standard deviation of operating income after depreciation (EBIT) over the past three years. Last, to account for the fact that firm financials might be depressed at the end of 1974 due to the 1973–1974 recession, we use three-year averages for the five firm-specific variables in the years 1972–1974 for the pre-period bonds.[22] Beyond these firm-specific characteristics, we also use the industry of the bond issuer for the matching. We then use the resulting coefficients (shown in Table A.1 in the appendix) to yield propensity scores, which we match to post-period firms.

[22] For the pre-period bonds that were rated between January and May 1975, we use the three averages of 1973–1975. We also tried four- and five-year averages; the sample size gets reduced, but the results are slightly higher in terms of economic magnitudes.

Table 6 Ratings Predictions Based on Moody's Pre-SEC-Regulation Ratings Model

Average rating for:	(1) Predicted rating	(2) Actual rating	(3) Difference	(4) *t*-stat.
Unmatched	2.891	3.202	0.311***	3.65
Matched	3.929	3.202	−0.727***	−2.42
No. of observations	1,264			

Notes. This table shows the average rating for the pre- and post-SEC-regulation-issued bonds for an unmatched and a matched sample of pre and post issues. A higher numerical rating indicates higher credit risk (worse letter rating). Unmatched refers to the average rating of all issues for the unmatched sample. Matched refers to a propensity score matched sample of pre issues where the propensity score was computed using firm size, leverage, pretax interest coverage, operating margin, and cash flow volatility as well as industry fixed effects as matching variables. For the matching, we use the three-year averages of these variables for the years 1972–1974 (1973–1975) for the pre-event issues that were rated until December 1974 (May 1975), and the values of the year in which the rating was assigned for the post-event issues. The matching was done using a one-to-one nearest neighbor matching with replacement.

***Indicates significance at the 1% level.

Using this model, we compare the rating for the post-period issue to what a bond of an issuer with similar size, financials, and industry classification would have been rated in the pre-period. Specifically, we propensity score match each post-period bond to a pre-period issue using the nearest neighborhood matching with replacement and compare their credit ratings by computing the average treatment effect of the treated.[23] Results are displayed in Table 6. The table shows that before matching, the average rating of the pre-period bonds was 2.891, while it was 3.202 for the post-period bonds. The difference between the two is highly significant. However, after the matching, the average predicted rating is 3.929. The difference is due to the treatment. Put differently, given a similar size, leverage, pretax interest coverage, operating margin, cash flow volatility, and industry, the predicted numerical rating for the post-period bonds is about 0.75 categories higher than the actual rating. The difference between the predicted and actual ratings is highly significant (*t*-statistic of −2.42). This result suggests that based on some of the most important firm-specific variables for the determination of a credit rating and the industry of the bond issuer, the post-period bond ratings are inflated by approximately three-quarters of a rating category. This result corroborates the previous findings where we artificially deflated/inflated the ratings and found slightly higher inflation magnitudes.[24]

[23] The results do not change if we use two, three, four or five control observations for the matching.

[24] These tests are similar to those in several previous papers that explored whether rating standards change over time (see Blume et al. 1998, Baghai et al. 2014, Alp 2013). Those papers implemented

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4.5. Robustness Tests

In this section, we provide three robustness tests to demonstrate that our main results hold when we change several key assumptions underlying our analyses. We first explore whether differing maturity structures between pre- and post-period bonds can explain our results. This is followed by tests that change the length of the evaluative window and by additional specifications that account for anticipatory effects of the regulatory change.

4.5.1. The Effect of Differing Maturities. One potential concern with our empirical approach relates to the maturity structure of the pre- versus post-period bonds. For the pre-period, we use all outstanding bonds as per the end of May 1975, while for the post-period bonds, we only use newly issued bonds. If a significant number of pre-period bonds have a shorter time to maturity, it could be argued that the earlier maturity dates of these bonds mechanically lead to higher default frequencies of the post-period bonds, given the same rating. We offer below several pieces of evidence to rule out this explanation as the main driver of the results presented above.

The average (median) maturity is 25 (25.04) years for the pre-period bonds and 22.45 (22.79) for the post-period bonds. If we compute the remaining time to maturity for the pre-period bonds as of May 1975, we get an average (median) remaining maturity of 19.37 (19.18) years. As expected, this is shorter than the one for the post-period bonds. However, the difference is not large in economic terms. Furthermore, the average pre-period bond matures after the end of our evaluative window. In fact, of all the pre-period bonds, only 19% mature before the end of the evaluative window, while the respective number is 11% for the post-period bonds.

One important aspect in this regard refers to when bonds actually default. For the pre-period bonds, we find that the average (median) time to default computed from May 1975 until the date of default is 8.06 (7.89) years, while it is 7.06 (7.10) for the post-period bonds. In the case of the pre-period bonds that defaulted in the evaluative window, the average (median) remaining time to maturity is 20.42 (17.97) years, and is 18.48 (20.05) years for the post-period bonds. This implies that the pre-period bonds defaulted, on average, after 39.47% of the outstanding time to maturity, while the post-period bonds did so after 38.20%. All these descriptive comparisons suggest that differing maturity structures between the two samples are not substantial enough to explain our results.

When we run regression specifications that include bond-maturity-related control variables in Table 7, we get results that are very close to the baseline estimates in Table 2. We run several specifications: first, we include the maturity from day of issuance until the maturity date (in days) as an additional control variable (column (I) of Table 7). Second, in column (II), we also include the squared term of the time to maturity in days to account for a potential nonlinear relationship between time to maturity and bond default. In column (III), we instead include the natural logarithm of the time to maturity in days to account for nonlinearities. In column (IV), we include a dummy variable indicating the year of maturity. Finally, in column (V), we include dummy variables for four-year maturity buckets to allow for a potential piecewise linear relationship between maturity and bond default.

The baseline point estimate in the specification with industry fixed effects (Table 2, column (II)), is 2.64 percentage points, and when we control for bond maturity in Table 7, the point estimates range from 2.34 to 2.64 percentage points, with the difference between the baseline point estimate being not statistically distinguishable from the point estimates of these specifications that include bond-maturity-related control variables. While statistical significance is somewhat smaller in these new regressions, it is caused by the reduction in sample size due to missing information about bond maturities. Overall, these tests show that differing maturity structures do not explain our main findings.

In a second formal test, we estimate duration models to address the potential concern that differing times to maturity across pre- and post-period-issued bonds might lead to censoring in the data. Duration models take into account several types of censoring of the data. In particular, they take into consideration the fact that bond issues may survive (and default) only after December 31, 1987 (right censoring). They also take into account the above mentioned fact that some bond issues defaulted or arrived to maturity (without defaulting) prior to December 31, 1987.

Specifically, the differing times to maturity are taken into account in the following way: For pre-period issues, we measure duration as the number of days between May 31, 1975 and the default date; between May 31, 1975, and December 31, 1987, for the pre-period issued bonds that do not default; and between May 31, 1975, and the maturity date for the nondefaulted bonds that come to maturity before

panel regressions with firm characteristics and year dummy variables. Since we are only interested in examining the post-period compared to the pre-period, individual year dummies are unnecessary in our setting. Furthermore, the propensity score matching approach we implement allows for a more precise matching of pre- and post-firms than simply controlling for firm financials (which those other papers could not implement given their panel approach). Otherwise, the approaches are the same in design.

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Table 7 Default Regressions Controlling for Bond Maturity

Defaults as of year-end 1987	(I)	(II)	(III)	(IV)	(V)
Post-dummy	0.0234*	0.0240*	0.0235*	0.0264*	0.0252*
	[0.0137]	[0.0137]	[0.0138]	[0.0135]	[0.0139]
Time to maturity	−0.0161	−0.0492			
	[0.0246]	[0.0613]			
Time to maturity squared		0.0151			
		[0.0344]			
Log (*time to maturity*)			−0.0056		
			[0.0091]		
Maturity year				0.0030***	
				[0.0010]	
Remaining maturity 4–8 *years*					0.0036
					[0.0071]
Remaining maturity 8–12 *years*					0.0591**
					[0.0298]
Remaining maturity 12–16 *years*					−0.0427**
					[0.0211]
Remaining maturity 16–20 *years*					0.0516
					[0.0323]
Remaining maturity >20 *years*					0.0180*
					[0.0099]
Observations	1,712	1,712	1,712	1,712	1,712
Adjusted R^2	0.144	0.144	0.143	0.144	0.150
Rating level fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes

Notes. This table presents results for our predictive default regressions. The dependent variable is an indicator variable taking on a value of one if a default of the bond occurs within our given evaluative window after the introduction of the NRSRO status. *Post-dummy* is an indicator variable for the post-NRSRO sample, i.e., for all new issues rated between July 1975 and December 1978. We include a rating dummy variable for each rating category and industry fixed effects in all regressions. In column (I), bond maturity is defined as the number of days from the day of issuance until the day of maturity; column (II) includes its squared term as well. In column (III), the bond maturity is defined as the natural log of the time to maturity in days. In column (IV), we include a dummy variable for the year of maturity. In column (V), we include dummy variables for four-year rating buckets till maturity (piecewise linear specification). Standard errors clustered at the issuer level are shown in brackets.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

December 1987. For the post-period bonds, we measure duration as the difference between the date of the assignment of the initial rating and the default date; between the date of the initial rating assignment and December 31, 1987, for the post-period-issued bonds that do not default; and between the date of the initial rating assignment and the maturity date for the nondefaulted bonds that come to maturity before December 1987.

Table 8 shows the results of the estimation of Cox proportional hazard models using different combinations of control variables. These results are consistent with our main results in Table 2, whereby post-rated bonds are significantly more likely to default after controlling for rating levels and the impact of censoring in the data due to differing bond maturities. We again conclude that bond maturity differences between pre- and post-period-issued bonds do not drive our results. Finally, it does not seem plausible that differing maturity structures can explain the findings for Z-score and rating downgrades. Overall, our results are consistent with Moody's inflating ratings after the regulatory change in June 1975.

4.5.2. Other Robustness Tests. First, in panel A of Table 9, we vary the length of the evaluative window. Our main results use a time window with an end date of December 1987, so we change the end date to both December 1986 and December 1988. Second, in panel B of Table 9, we remove any ratings that were assigned in the five months preceding the regulatory change, from January to May 1975 (the introduction of the NRSRO status took place in June 1975), while at the same time varying the length of the evaluative period. Although the SEC actions were to a large degree unexpected (see the description of the "no action" letter process in Section 2), this test accounts for the possibility of any anticipation of the regulatory change. As Table 9 displays, in all of these variations, the results of our main analysis remain unchanged.

5. Alternate Explanations

With any natural experiment, it is important to show that our results are specific to the event under study

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Table 8 Default Regressions—Cox Proportional Hazard Model

Default until 1987:	(1)	(2)	(3)	(4)
Post-dummy	2.3978***	2.4594***	3.4304***	3.2579***
	[0.6554]	[0.7119]	[1.1782]	[1.1383]
Asset size (million USD)			0.2326	5.6829
			[4.7416]	[90.538]
Leverage			6.5855	22.718**
			[10.163]	[35.792]
Interest coverage			9.8377	1.0212
			[76.459]	[9.3652]
Operating margin			0.6867	1.5101
			[2.3093]	[3.0945]
Observations	1,712	1,712	1,137	1,137
Pseudo-R^2	0.144	0.178	0.144	0.191
Rating level fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	No	Yes	No	Yes

Notes. The table shows hazard ratios from the estimation of a Cox proportional hazard model. For the pre-period issues, the duration is computed as the difference in days between the default date and May 31, 1975; between December 31, 1987, and May 31, 1975, in the case of no default; and between May 31, 1975, and the date of maturity in the case of a maturity before December 31, 1987. For the post-period issues, the duration is computed as the default date and the date of the initial rating; between December 31, 1987, and the date of the first rating in the case of no default; and between the date of the initial rating and the maturity date in the case of a maturity before December 31, 1987. The estimation is performed on the subset of bonds with data on time to maturity (same subset of bonds as the one shown in Table 7). All other variables are defined in Table 2 and included as indicated in the table. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

** and *** indicate significance at the 5% and 1% levels, respectively.

and not driven by time trends or some other confounding factor. This section provides several tests designed to help rule out competing explanations for our main findings.

Table 9 Default Regressions—Robustness Tests

Defaults per end of:	1986	1987	1988
Panel A: Different time periods			
Post-dummy	0.0219**	0.0264**	0.0264**
	[0.0111]	[0.0116]	[0.0121]
Observations	2,496	2,496	2,496
Adjusted R^2	0.122	0.121	0.115
Rating level fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Panel B: Anticipation of event			
Post-dummy	0.0240**	0.0289**	0.0280**
	[0.0120]	[0.0125]	[0.0130]
Observations	2,351	2,351	2,351
Adjusted R^2	0.126	0.126	0.120
Rating level fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Notes. This table presents several alternative specifications for our main predictive default regressions. All variables are defined in Table 2. In panel A, we vary the time windows over which we measure default frequencies. Our baseline window in Table 2 ends in December 1987; we shorten the time window by one year in the first column, and we lengthen it by one year in the third column. In panel B, we also end the pre-SEC-certification sample period in December 1974; i.e., we exclude observations between January 1975 and May 1975 to account for the potential anticipation of the event. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

**Indicates significance at the 5% level.

5.1. Macro Events

The tests in this section are designed to rule out that macro events drive our results. The pre-period for our analysis includes a recession lasting from November 1973 to March 1975. An oil shock occurred in 1973, whereby oil prices rose from approximately \$3 to \$12 per barrel, a shock that is largely blamed for the ensuing recession. The post-period for our results also includes a recession in 1981–1982, but this occurs after any of the ratings in our tests are determined. Our tests also use the same period for the outcomes for both treatment and controls, so to the extent any macro events occur after ratings are measured, they affect both samples equally. So any macro changes in the post-period should not compromise our results.

Furthermore, as pointed out previously, rating agencies state that they "rate through the cycle." If rating agencies do not adjust rating levels because of normal ebbs and flows of the business cycle, then a bond rated in a recession will receive the same rating as if it were rated in an expansion. If the timing of when a bond is rated does not impact default risk, this will have no impact on our results. If, however, bonds rated in a recession have a subsequently higher level of default risk, then ratings will appear inflated when given during recessions. As the 1973–1975 recession falls into the pre-SEC-certification period, this should introduce a bias against finding ratings inflation in the post-period (the opposite of what we find). Additionally, our results in the cross section are difficult to reconcile with business cycle effects. For example,

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Table 10 Test for Macro Trends I: Excluding Recession Years

Excluded time period:	None	1973	1974	November 1973–March 1975
		Panel A: Defaults		
Post-dummy	0.0264**	0.0292**	0.0295**	0.0340**
	[0.0116]	[0.0121]	[0.0127]	[0.0142]
Observations	2,496	2,368	2,262	2,138
Adjusted R^2	0.121	0.123	0.121	0.132
Rating level fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
		Panel B: Indicator for drop in Z-score levels		
Post-dummy	0.1446***	0.1505***	0.1536***	0.1401***
	[0.0526]	[0.0528]	[0.0530]	[0.0532]
Starting Z-score	0.1716***	0.1721***	0.1693***	0.1863***
	[0.0230]	[0.0232]	[0.0228]	[0.0239]
Observations	1,066	1,031	994	942
Adjusted R^2	0.203	0.202	0.205	0.222
Rating level fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
		Panel C: Downgrades to SG		
Post-dummy	0.0777***	0.0764***	0.0861***	0.0807***
	[0.0273]	[0.0280]	[0.0290]	[0.0309]
Observations	2,219	2,107	1,987	1,865
Adjusted R^2	0.084	0.077	0.088	0.092
Rating level fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes

Notes. This table shows results for our predictive default (panel A), drop in Z-score levels (panel B), and downgrades to SG (panel C) regressions, whereby we exclude specific time periods related to the 1973–1974 recession from our pre-SEC-certification period. The first column displays our baseline results. In the second column, we remove all assigned ratings in 1973. In the third column, we remove all assigned ratings in 1974. In the fourth column, we remove all assigned ratings between November 1973 and March 1975, which corresponds to the NBER-defined recession period. All variables are defined in Tables 2–4. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

** and *** indicate significance at the 5% and 1% levels, respectively.

changes in the business cycle do not directly imply that Baa firms in particular should receive inflated ratings in the post-period. Nevertheless, we conduct several additional tests in this section to rule out that macro events in 1973–1975 are compromising our conclusions.

Our first set of tests alters the years used for analysis in the pre-period. Specifically, we remove any bonds that received a rating during the recession of 1973–1975. We conduct separate tests excluding all bonds with newly assigned ratings during 1973, 1974, and during the recession ranging from November 1973 to March 1975, the period which was defined by the National Bureau of Economic Research (NBER) as the recession period. Table 10 shows that our main results (reported for reference in the first column) are unchanged and even slightly stronger for all three main outcome variables when we estimate our tests while removing the rating assignments made during the 1973–1975 recession period in the second, third, and fourth columns. We conclude from this table that our main results are not driven by the ratings assigned during the 1973–1975 recession period.

Next, we perform a placebo or falsification test on a rolling window spanning from 1970 to 1980. This test allows us to rule out time trend effects surrounding our event window. We implement this test by running the same specification as in our main test with the only difference being that instead of taking June 1975 as the event month, we use different placebo event years. Table 11 shows the results of these tests. In column (1), we set the placebo event month to be June 1970. Specifically, we record all outstanding bond ratings per end of May 1970 and all newly assigned bond ratings between July 1970 and December 1973. We then record the realized defaults until December 1982, i.e., nine years after the end of the placebo post-period. This test is identical to the baseline default analysis, with the difference being that there was no regulatory event in June 1970. Consequently, we should not find any difference in the default likelihood of the placebo pre- and post-rated bonds if our results are driven by the SEC actions. In column (2) we set the placebo event month to be June 1971 and repeat this exercise until we finally set the placebo event month to June 1980 in column (11).

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Table 11 Test for Macro Trends II: Rolling Window Placebo Tests

	(1) 1970	(2) 1971	(3) 1972	(4) 1973	(5) 1974	(6) 1975	(7) 1976	(8) 1977	(9) 1978	(10) 1979	(11) 1980
Post-dummy	0.0023	0.0114	0.0099	0.0116	0.0131	0.0264**	0.0361***	0.0169	0.0057	0.0007	0.0045
	[0.0062]	[0.0097]	[0.0084]	[0.0113]	[0.0114]	[0.0116]	[0.0137]	[0.0127]	[0.0118]	[0.0121]	[0.0130]
Observations	1,650	1,854	2,102	2,264	2,397	2,496	2,663	2,884	3,094	3,233	3,215
Adjusted R^2	0.295	0.101	0.069	0.032	0.081	0.119	0.117	0.125	0.148	0.156	0.156
Rating level fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table presents results for a rolling window of placebo (falsification) tests for our predictive default regressions. The dependent variables and all control variables are defined in Table 2. We replicate the estimation of specification (II) of Table 2 with 10 other placebo events. Specifically, we shift the event window by one year for every column, starting with 1970 in column (1) to 1980 in column (11). We shade column (6) because it represents the actual event window (see Table 2, panel A, column (II)). The event is in June 1975. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

** and *** indicate significance at the 5% and 1% levels, respectively.

Column (6) shows our baseline result from Table 2 (column (II)) where June 1975 is the event month. As in our main specification, we control for industry and credit rating level fixed effects.

The pattern we observe from Table 11 highlights the lack of any clear trend both before and after the true event window. The table shows no significance in any placebo event year leading up to 1975. Column (6) is shaded and represents the results from our main specification in Table 2 using the true event window. It is the only year where we find significance with the exception of 1976, the year following the event. The significance obtained for 1976 can be explained by a potentially delayed reaction of Moody's to the regulatory change. By shifting the event window by just one year, there is only a slight misclassification of the pre-period bonds that include all bonds from June 1975 to May 1976 to the pre-period sample. These results would be consistent with Moody's only gradually inflating ratings after June 1975. Once the window is shifted by two or more years, the results disappear. The results in Table 11 reject any concern for time trends affecting our results. We conclude instead that a structural break in 1975 was triggered by the new SEC regulations.

Our last placebo test makes use of the recession of 1981–1982. In Table 12, we replicate all the estimations of Table 2, panel A; however, we set the placebo event month to be June 1983 instead of June 1975. In doing so, we replicate the macro environment of our main test by also having a major recession in the two years prior to the event year, i.e., the recession years 1981 and 1982. The 1981–1982 recession is comparable in many ways to the 1973–1975 recession. Both represent two of the most severe recessions since the Great Depression, and both lasted approximately 16 months. Unemployment reached as high as 11% in the 1981–1982 recession and 9.0% in the 1973–1975 recession, and both recessions saw gross domestic product declines of approximately 3% peak to trough. Finally, both recessions were preceded by oil shocks, with the Iranian revolution in 1979 preceding the 1981–1982 recession and the OPEC (Organization of the Petroleum Exporting Countries) oil embargo of 1973 preceding the 1973–1975 recession.

The results do not show a significantly positive coefficient on *post-dummy* in any of the specifications. If anything, the coefficient is negative (albeit not significant) in all specifications. The lack of results in Table 12 supports that our main results are not driven

Table 12 Test for Macro Trends III: 1981–1982 Recession

Replication of panel A of Table 2 with a placebo event year of 1983

	(I)	(II)	(III)	(IV)
Post-dummy	−0.0019	−0.0077	−0.0162	−0.0130
	[0.0092]	[0.0095]	[0.0110]	[0.0090]
Asset size (million USD)			−0.0689	−0.0463
			[0.1436]	[0.1439]
Leverage			−0.0401	−0.0522
			[0.0929]	[0.1014]
Interest coverage			−1.3426*	−1.3907**
			[0.6882]	[0.7013]
Operating margin			−0.0177	−0.0109
			[0.0473]	[0.0470]
Observations	6,054	6,054	3,364	3,364
Adjusted R^2	0.204	0.220	0.167	0.167
Rating fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	No	Yes	No	Yes

Notes. This table presents results for placebo (falsification) tests for our predictive default regressions. The dependent variables and all control variables are defined in Table 2. We replicate the estimations of panel A in Table 2; however, we set the placebo event year to be 1983 instead of 1975. In doing so, we replicate the macro environment of our main test by also having a major recession in the two years prior to the event year, but the 1983 event year does not have the corresponding NRSRO introduction. If the recession confounds our results, we expect *post-dummy* to load positively as in our baseline regression results in Table 2. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

* and ** indicate significance at the 10% and 5% levels, respectively.

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by the macro environment at the time of the regulatory change under study.[25]

Taken together, the results from the tests in this section show that the ratings inflation documented in this paper is not due to some other confounding event or omitted macro factor during this time period.

5.2. S&P's 1974 Change in Payment Method

A significant area of ratings research looks at the differences in ratings quality between issuer pay and investor pay, with the overall conclusion that the issuer-pay model lowers rating quality (see Beaver et al. 2006, Jiang et al. 2012, Cornaggia and Cornaggia 2013, Bruno et al. 2016, Xia 2014). Bruno et al. (2016) find that when an investor-paid rating agency (Egan Jones) becomes an NRSRO, its ratings quality does not decline.

Jiang et al. (2012) show that the switch of S&P's business model from investor pay to issuer pay introduced a bias in S&P's ratings. They argue that after switching to issuer pay, S&P assigned relatively higher ratings to attract more business and generate higher returns. The switch in payment model took place around June/July 1974, a time period that is reasonably close to the introduction of the NRSRO status in June 1975. As such, our findings could be related to this event. Specifically, if Moody's were to inflate its ratings in reaction to S&P's change in business practices, it could potentially explain why default frequencies in a given rating class increased. However, we show in this section that our findings are unlikely to be driven by this confounding explanation.

To show that our results are not driven by S&P's switch to issuer pay, we design the following test. We focus only on ratings of bonds issued in the pre–SEC-certification period, i.e., until May 1975. We then use the switch of S&P's payment model in June/July 1974 as the event month and define the dummy variable *post-dummy* to take on a value of one if the bond was rated between July 1974 and May 1975 and zero if it was rated before July 1974. Focusing on observations until May 1975 enables us to isolate the effect of a potential reaction of Moody's to S&P's payment model switch from the SEC certification event we focus on in this study.[26] We then run the predictive tests for all three main outcome variables. Other than redefining the post-event dummy, the regression setup is the same as before. If Moody's reacted to inflated S&P ratings after the switch of the payment model, the coefficient of *post-dummy* should load positively and significantly in these regressions. Table 13 contains the results.

Table 13 Is It a Reaction to S&P's Switch to Issuer Pay?

Dependent variable:	Defaults	Indicator for drop in Z-score levels	Downgrades to SG
Post-dummy	0.0010	0.0073	−0.0375
	[0.0122]	[0.0536]	[0.0255]
Starting Z-Score		0.1685***	
		[0.0250]	
Observations	1,995	907	1,809
Adjusted R^2	0.056	0.169	0.062
Rating level fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes

Notes. This table shows results for predictive defaults, drop in Z-score levels and downgrades to SG regressions over a sample window corresponding to a potential reaction of Moody's to S&P's switch to issuer pay. Specifically, the window of analysis is restricted to observations prior to May 1975 to isolate the change of S&P's payment model in June 1974 from the change in SEC's ratings regulation in June 1975 (introduction of NRSRO). The newly defined *post-dummy* variable takes a value of one for all new issues over the period from July 1974 to May 1975 and zero otherwise. All other variables are defined in Tables 2–4. To save space, we do not show coefficients for the rating level fixed effects. Standard errors clustered at the issuer level are shown in brackets.

***Indicates significance at the 1% level.

From the first column of Table 13, we can see that the coefficient of *post-dummy* is not significant and close to zero, implying that default frequencies of bonds rated before July 1974 and those rated between July 1974 and May 1975 did not differ significantly regardless of the window over which we record defaults. In the second column, we run the same regressions using the indicator of a drop in Altman Z-score as the dependent variable. This time, *post-dummy* loads positively and is again not statistically significant. Finally, in the third column, we use the indicator of a downgrade to speculative grade as the dependent variable. The coefficient of *post-dummy* is again negative and not significant. In all, these results suggest that the inflation in Moody's ratings was not caused by a reaction to S&P's switch to the issuer-pay model.[27]

[25] We run the same placebo test for the Z-score tests (not tabulated here). The results show no similarities to those run on our 1975 event window and hence also provide evidence supporting that our results are not driven by the 1973–1974 recession period.

[26] In the placebo post-period from July 1974 to May 1975, there are 276 bond ratings. This represents 14% of the sample that includes 1,719 bond ratings from the placebo pre-period up to May 1974. We believe that this is a sufficiently large sample size to be able to draw statistical inference from this test.

[27] In untabulated results, we also examined whether stock market investors reacted differently to rating changes following SEC certification. The stock market reaction to rating changes is not statistically different in the post-SEC-certification period. While several studies show that stock prices generally react to credit rating changes (e.g., Holthausen and Leftwich 1986, Hand et al. 1992, Goh and Ederington 1993), the impact of the SEC certification could arguably impact returns either direction. On the one hand, if SEC certification provided more power to rating agencies in the marketplace, we could expect that announcement returns would increase in the post-period as investors rely more on these ratings. On the

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As an additional diagnostic, we make use of Moody's change to investor pay in October 1970. We conduct tests similar to the main tests of this paper around that date, and we find no evidence of rating inflation following this date;[28] that is, Moody's does not inflate ratings after it changes to issuer pay in October 1970.

5.3. Ratings Accuracy

An alternative implication of the regulatory change is that ratings accuracy would decrease after SEC certification. Rating agencies might rationally exert less effort to rate firms or, because of the increased demand for ratings following the regulations, ratings analysts may not be able to spend as much time on determining ratings. This implies that ratings would become more random in the post-period as opposed to deliberately inflated. We examine this alternative in two tests. First, we examine the ordinal integrity of ratings in the pre-period versus the post-period. We examine the default likelihood for each rating category to see whether the ordinal ranking for ratings is preserved in the post-period. We find that the ordering of ratings is maintained in the post-period, such that each lower rating category shows a higher probability of default than the rating category above it. For example, a Baa-rated firm in the post-period is 3.6 times more likely to default than an A-rated firm, and a B-rated firm is 3.1 times more likely to default than a Ba firm. For pre-period-rated firms, these ratios are equal to 3.0 and 4.7, respectively.

Our second test examines the likelihood of subsequent upgrades and downgrades following the initial rating.[29] If ratings are initially less accurate, we expect rating agencies to change ratings more frequently going forward. However, if ratings are inflated, we expect a differential impact on the likelihood of downgrades relative to upgrades. The evidence in Table 14 is consistent with ratings inflation. From panel A, we see that the likelihood of a subsequent downgrade increases by 5% (from 14% to 19%) and is statistically significant at the 1% level, but there is no significant difference for upgrades. Importantly, results from panel B show that the effect is even stronger for Baa firms; in particular, the likelihood of a subsequent downgrade increases by 9% percentage points (from 13% to 22%) and is statistically significant at the 5% level. We again find no significant difference for upgrades. This is consistent with our earlier findings that ratings inflation around the investment grade threshold seems to be particularly valuable for bond issuers.

Table 14 Frequency of First Rating Changes

	Downgrades	Upgrades
	Panel A: All bonds	
Post-dummy	0.0507***	−0.0009
	[0.0181]	[0.0145]
Observations	2,466	2,466
Adjusted R^2	0.023	0.051
Rating level fixed effects	Yes	Yes
	Panel B: Baa-rated bonds only	
Post-dummy	0.0921**	−0.0567
	[0.0467]	[0.0495]
Observations	395	395
Adjusted R^2	0.007	0.001

Notes. This table displays results for differences in frequency of the first rating change (upgrade or downgrade) for issues rated in the pre-SEC-certification period relative to issues rated in the post-SEC-certification period. The evaluative period over which upgrades and downgrades are measured runs from July 1975 to March 1982 for pre-SEC-certification-rated bonds, and from issuance to March 1982 for post-SEC-certification-rated bonds issued between July 1975 and December 1978. The evaluative period ends in March 1982 just before the introduction of the notching system by Moody's. In panel A, we report the regression results for all bonds, whereby we control for rating level fixed effects and use a post-period dummy variable to test for differences in the likelihood of being downgraded (first column) or upgraded (second column) between pre-period and post-period bond issues. In panel B, we consider only issues that are initially rated Baa (above investment grade cutoff).

** and *** indicate significance at the 5% and 1% levels, respectively.

Ratings inflation benefits institutional investors that are constrained by regulations, but losing accuracy in general would make ratings less attractive for non-constrained investors. Moody's appears to recognize this by having maintained accuracy generally while inflating ratings levels.

6. Conclusion

Two of the key institutional features of the credit rating industry have been the widespread use of ratings in rules and regulations and the restrictions placed on competition caused by the NRSRO designation system. Understanding how these regulations impact credit ratings is essential for addressing concerns about the current rating system. In this paper, we examine one of the watershed events in the history of credit ratings, the introduction of the NRSRO status, and its use in regulations. June 26, 1975, marks the launching date for the significant expansion of

other hand, investors could anticipate a lower quality of credit ratings following the SEC certification and would therefore react less to credit rating changes.

[28] When October 1970 is used as the placebo event month, we obtain an insignificant coefficient of 0.0068 for *post-dummy*.

[29] We end the evaluative period for this test in March 1982 because Moody's introduced the notching system in April 1982, which could introduce a bias in the analysis; undoubtedly more upgrades and downgrades occur once ratings become finer.

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ratings use in rules and regulations, and the establishment of significant barriers to entry in the ratings industry.

The evidence we present in this paper is consistent with the regulatory changes instituted in June 1975 leading to ratings inflation. For a given rating, a bond prior to June 1975 was less likely to default, less likely to experience a deterioration in overall financials, and less likely to be downgraded to speculative grade than a bond with the same rating given after the regulation change. We conduct several diagnostic tests to rule out potential confounding explanations for these results, including differing macro environments at the time.

Our results suggest that the SEC certification led to ratings inflation, but the rank ordering of ratings seems to have been preserved throughout (i.e., a Baa bond remains a worse credit risk than a Ba bond, e.g., in the post-period). We also present evidence that suggests that ratings inflation is more pronounced around the investment grade cutoff.

Our paper speaks directly to the impact of competition and the use of ratings in rules and regulations on ratings quality. Some recent papers indicate that more competition can have a detrimental impact on ratings quality (e.g., Becker and Milbourn 2011, Bolton et al. 2012). Our evidence indicates that the joint effect of restricting competition and increasing the importance of ratings for rules and regulations can also have a detrimental effect on ratings quality through ratings inflation. By increasing barriers to entry and providing a captive customer base for rating agencies, the new regulations of 1975 changed the incentive structure for rating agencies, which led to a material upward bias in ratings. Any change to the ratings landscape today should consider the nature of the incentives created by the regulatory environment initiated at that time.

Acknowledgments

An earlier version of this paper circulated under the title "Did Government Regulations Lower Credit Rating Quality?" The authors thank an anonymous referee, an anonymous associate editor, Amit Seru (the department editor), Aysun Alp, Murillo Campello, Jess Cornaggia, Kimberly Cornaggia, Gunter Löffler, Marcus Opp (American Finance Association discussant), Michael Roberts, and Phil Strahan, as well as seminar participants at the American Finance Association meeting 2015, American University, Boston College, Bristol University, Exeter University, George Mason University, Georgetown University, Sabanci University, and the University of Lugano for helpful comments. The authors also thank Felipe Restrepo Gomez and Saeid Hoseinzade for their research assistance. All errors are the authors' responsibility.

Appendix

Table A.1 First-Stage Probit Estimation Results

Dependent variable:	*Numerical rating*
Asset size (million USD)	−11.2973
	[8.8184]
Leverage	3.9333***
	[0.4503]
Interest coverage	3.4547*
	[2.0911]
Operating margin	0.9835***
	[0.3590]
Cash flow volatility	−0.0002
	[0.0006]
Observations	1,264
Pseudo-R^2	0.205
Matched on industry	Yes

Notes. This table displays results of the first-stage probit estimation with the numerical rating level as the dependent variable. We use three-year averages of the years 1972–1974 (1973–1975) for the bonds that were issued until December 1974 (May 1975). To save space, we do not report the coefficients for the industries used in the matching.

* and *** indicate significance at the 10% and 1% levels, respectively.

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